南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course Name: General physics B (II) / A **Dept:** PHYSICS

Exam Duration: 2 hours **Exam Paper Setter:** Physics teaching team

Question No.	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8		
Score	30	6	10	10	10	10	12	12		

This exam paper contains 7 questions and the score is 100 in total. (Please hand in your exam paper, answer sheet, and your scrap paper to the proctor when the exam ends.)

Q1. Multiple Choice Questions (3 points each), there is only one correct answer for each question.

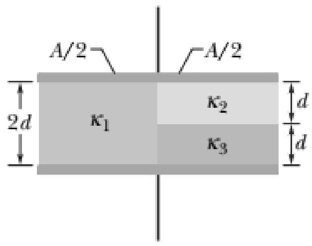
1-5) CBCCD 6-10) CCACD

Q2. Blank-filling Questions. (Just write down the result, and the detailed process is not needed.)

1. **(3 points)** 636.50-637.50V
2. **(3 points)** 0.59-0.60nC

Text Questions: (Please write down the detailed process.)

Q3. (10 points)



Let

$$C_1 = \varepsilon_0(A/2)\kappa_1/2d\ .$$
$$C_2 = \varepsilon_0(A/2)\kappa_2/d\quad ,$$
$$C_3 = \varepsilon_0(A/2)\kappa_3/d\ ,$$

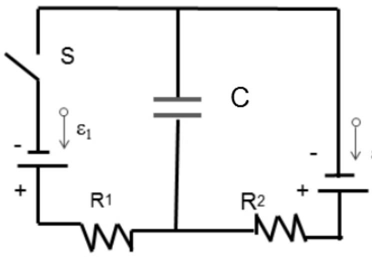
Note that C_2 and C_3 are effectively connected in series, while C_1 is effectively connected in parallel with the C_2-C_3 combination. Thus,

$$C = C_1 + \frac{C_2 C_3}{C_2 + C_3} = \frac{\varepsilon_0 A}{4d}(\kappa_1 + \frac{2\kappa_2\kappa_3}{\kappa_2 + \kappa_3})$$

With $A = 10.5\times10^{-4}\text{m}^2$, $d = 3.56\times10^{-3}\text{m}$, $\kappa_1 = 21.0$, $\kappa_2 = 42.0$ and $\kappa_3 = 58.0$, we find the capacitance to be

$$C = \frac{(8.85\times10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2)(10.5\times10^{-4}\ \text{m}^2)}{4(3.56\times10^{-3}\ \text{m})}\left(21.0 + \frac{2(42.0)(58.0)}{42.0+58.0}\right) = 4.55\times10^{-11}\text{F}.$$

Q4. (10 points)



When S is open for a long time, the charge on C is $q_i = \varepsilon_2 C$. When S is closed for a long time, the current i in R_1 and R_2 is

$$i = (\varepsilon_2 - \varepsilon_1)/(R_1 + R_2) = (3.0\ \text{V} - 1.0\ \text{V})/(0.30\ \Omega + 0.40\ \Omega) = 2.86\ \text{A}.$$

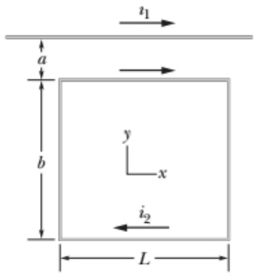
The voltage difference V across the capacitor is then

$$V = \varepsilon_2 - \quad iR_2 = 3.0\ \text{V} - (2.86\ \text{A})\ (0.40\ \Omega) = 1.87\ \text{V}.$$

Thus the final charge on C is $q_f = VC$. So the change in the charge on the capacitor is

$$\Delta U = U_f - U_i = 1/2(V^2 - \varepsilon_2^2)\text{C} = 1/2\ (1.87^2\text{V} - 3.0^2\ \text{V})\ (15\ \mu\text{F}) = -41.3\ \mu\text{J}.$$

Q5. (10 points)



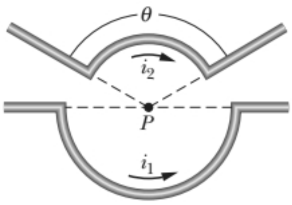
The magnitudes of the forces on the sides of the rectangle that are parallel to the long straight wire (with i_1 = 30.0 A) are computed using Eq. 29-13, but the force on each of the sides lying perpendicular to it (along our y axis, with the origin at the top wire and $+y$ downward) would be figured by integrating as follows:

Fortunately, these forces on the two perpendicular sides of length b cancel out. For the remaining two (parallel) sides of length L, we obtain

$$F=\frac{\mu_0 i_1 i_2 L}{2\pi}\left(\frac{1}{a}-\frac{1}{a+b}\right)=\frac{\mu_0 i_1 i_2 L b}{2\pi a(a+b)}$$

and $\vec{F}$ points toward the wire, or $+\hat{\mathrm{j}}$. That is, $\vec{F}=(3.20\times10^{-3}\,\mathrm{N})\hat{\mathrm{j}}$ in unit-vector notation.

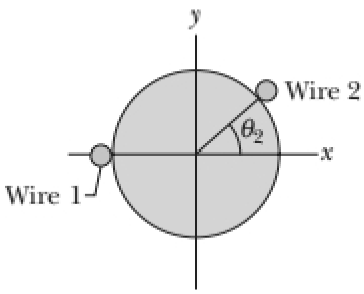
Q6. (10 points)



The radial segments do not contribute to magnetic field at P, If designates $\hat{\boldsymbol{k}}$ the direction "out of the page" , then

$$\begin{aligned}\vec{B}&=\frac{\mu_0\, i_1\phi_1}{4\pi r_1}\hat{\mathrm{k}}-\frac{\mu_0\, i_2\phi_1}{4\pi r_2}\hat{\mathrm{k}}\\ &=\frac{\mu_0\left(0.40\,\mathrm{A}\right)\left(\pi\,\mathrm{rad}\right)}{4\pi\left(0.050\,\mathrm{m}\right)}\hat{\mathrm{k}}-\frac{\mu_0\left(0.8\mathrm{A}\right)\left(2\pi/3\,\mathrm{rad}\right)}{4\pi\left(0.040\,\mathrm{m}\right)}\hat{\mathrm{k}}=-(1.68\times10^{-6}\,\mathrm{T})\end{aligned}$$

Q7. (12 points)



By the right-hand rule (which is "built-into" Eq. 29-3) the field caused by wire 1's current, evaluated at the coordinate origin, is along the $+y$ axis. Its magnitude B_1 is given by Eq. 29-4. The field caused by wire 2's current will generally have both an x and a y component, which are related to its magnitude B_2 (given by Eq. 29-4), and sines and cosines of some angle. A little trig (and the use of the right-hand rule) leads us to conclude that when wire 2 is at angle θ_2 (shown in Fig. 29-61) then its components are

$$B_{2x} = B_2 \sin\theta_2, \quad B_{2y} = -B_2 \cos\theta_2 .$$

The magnitude-squared of their net field is then (by Pythagoras' theorem) the sum of the square of their net x-component and the square of their net y-component:

$$B^2 = (B_2 \sin\theta_2)^2 + (B_1 - B_2 \cos\theta_2)^2 = B_1^2 + B_2^2 - 2B_1B_2 \cos\theta_2 .$$

(since $\sin^2\theta + \cos^2\theta = 1$), which we could also have gotten directly by using the law of cosines. We have

$$B_1 = \frac{\mu_0 i_1}{2\pi R} = 50 \text{ nT}, \quad B_2 = \frac{\mu_0 i_2}{2\pi R} = 30 \text{ nT}.$$

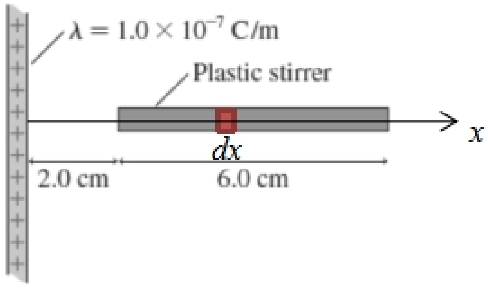
so

$$43.59^2 = 50^2 + 30^2 - 2\times 50\times 30\cos\theta_2 .$$

$$\cos\theta=0.5$$

$$\theta=60^\circ$$

Q8. (12 points)



For the infinite long, charged wire, the electric field around it is

$$E = \frac{\lambda}{2\pi\varepsilon_0 r}$$

r is the distance to the wire,so the net electric force on the plastic stirrer with 10nC charge on it is

$$dF = Edq = \frac{\lambda}{2\pi\varepsilon_0 x}\frac{Q}{l}dx$$

$$F = \int_d^{d+l} \frac{\lambda}{2\pi\varepsilon_0 x}\frac{Q}{l}dx = \frac{\lambda Q}{2\pi\varepsilon_0 l}\ln\frac{d+l}{d} = \frac{1.0\times 10^{-7}\times 10\times 10^{-9}}{2\pi\times 8.85\times 10^{-12}\times 0.06}\ln\frac{0.02+0.06}{0.02}(N) = 4.16\times 10^{-4}N$$